Healthcare

Tittle: Healthcare Patient Data

Topics:DDL, DML, FIFO, LIFO, Multithreading, Error handling

Problem Statement:

Multithreading, LIFO, FIFO,DDL, DML, Error handling this topics are integrated in the healthcare patients data explain with code example

Roles/Responsibility:

Database designer: DDL/DML

- Create database structure
- Insert/retrieve data

System Analyst: LIFO/FIFO

- Manage emergency cases
- Manage OPD queue

Developer: Multithreading

- Handle parallel tasks

Tester: Error handling

- Prevent program crash
- Validate system behavior

Code Review-Topics Integrated:

√ Create table-DDL-Define Patient Schema

√ Queue –FIFO-Patient queue

√

Complete Code With Professional Comments:

```
 # ===================== DDL (Data Definition Language)
=====================

# DDL is used to define database structure (tables, columns)


"""

CREATE TABLE patients (          -- Create a table named patients

    patient_id INT,              -- Stores unique patient ID (number)

    name TEXT,                   -- Stores patient name

    disease TEXT,                -- Stores disease name

    visit_type TEXT              -- Stores visit type: Inpatient / Outpatient

);

"""


# ===================== DML (Data Manipulation Language)
=====================

# DML is used to insert, update, delete data in tables


"""

INSERT INTO patients VALUES (     -- Insert a new record into patients table

    1,                           -- patient_id value
```

```
    'Ramesh',                  -- patient name
    'Diabetes',                -- disease name
    'Inpatient'                -- visit type
);
"""


# ===================== FIFO (First In First Out)
=====================
# Queue is used where first patient comes first served (OP queue)


from queue import Queue         # Import Queue class for FIFO operations


op_queue = Queue()              # Create an empty outpatient queue


op_queue.put("Patient-101")      # Add first OP patient to queue
op_queue.put("Patient-102")      # Add second OP patient to queue
op_queue.put("Patient-103")      # Add third OP patient to queue


print("Next OP Patient:", op_queue.get())
# get() removes and returns the first patient (FIFO)


# ===================== LIFO (Last In First Out)
=====================
```

```
# Stack is used for emergency where last critical case handled first

emergency_stack = []            # Create empty stack using list

emergency_stack.append("Accident Case")
# Add accident case to stack

emergency_stack.append("Heart Attack")
# Add heart attack case on top of stack

print("Immediate Attention Patient:", emergency_stack.pop())
# pop() removes last added patient (LIFO)

# ===================== Multithreading =====================
# Multithreading allows multiple hospital tasks to run simultaneously

import threading                # Import threading module

def monitor_vitals():           # Function to monitor patient vitals
    print("Monitoring patient vitals in ICU")

def generate_report():          # Function to generate lab report
```

```
    print("Generating lab report for patient")

t1= threading.Thread(target=monitor_vitals)
# Create thread for monitoring vitals

t2 = threading.Thread(target=generate_report)
# Create thread for report generation

t1.start()                    # Start vitals monitoring thread
t2.start()                    # Start lab report thread

# ==================== Error Handling ====================
# Error handling prevents system crash due to wrong input

try:
    patient_id = int("ABC")      # Invalid input (string instead of number)
    print("Patient ID:", patient_id)
except ValueError:              # Catches conversion error
    print("Error: Invalid Patient ID entered – system protected")

# ==================== Integrated Healthcare Flow
====================
# Checking remaining patients in OP queue and emergency stack
```

```
if not op_queue.empty():        # Check if OP queue still has patients
    print("Remaining OP handled")


if emergency_stack:             # Check if emergency stack is not empty
    print("Other emergency cases pending")
```

Expected Output:

Next OP Patient: Patient-101

Immediate Attention Patient: Heart Attack

Monitoring patient vitals in ICU

Generating lab report for patient

Error: Invalid Patient ID entered – system protected

Remaining OP handled

Other emergency cases pending

Interview Questions:

- √ DDL is used for
  - o Structure
- √ Command used to create table
  - o Create
- √ DML is used for
  - o Data
- √ Command to insert records

  - Insert
- √ FIFO full form
  - First in first out
- √ Data structure used in op system
  - Queue
- √ LIFO full form
  - Lastin first out
- √ Data structure used in emergency cases
  - Stack
- √ Multithreading meaning
  - Concurrency
- √ Python module for multithreading
  - Threading
- √ Purpose of error handling
  - Safety
- √ Error raised for invalid input type
  - Value Error